# Supplementary material: The Hubble Tension: A Conceptual Summary from a Session Without Computation or Literature Search

HEP-Genesis-Agent session record
Project: ramp-features

2026-10-17

*Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.*

## A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

### A.1 Session

- Source: cli client; conversation `1dffc244`
- Started: 2026-10-15 22:42:59; report generated: 2026-10-17 17:02:53
- Project: ramp-features (`/Users/nramachandra/hep-genesis-testing/projects/ramp-features`)
- Compute execution site(s) selected in the session: local
- 3 user turn(s), 3 tool call(s) (0 returned errors), 0 figure(s), 0 data file(s)

### A.2 Models

| Stage | Model | Endpoint |
|---|---|---|
| quick | `claude-opus-5-5` | `https://mag.genesis.americansciencecloud.org/v1` |
| report | `claude-opus-5-5` | `https://mag.genesis.americansciencecloud.org/v1` |
| coder | `claude-opus-5-5` | `https://mag.genesis.americansciencecloud.org/v1` |

### A.3 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

| # | Tool | Arguments | Outcome |
|---|---|---|---|
| 1 | genesis write_report | {"focus": "Conceptual explanation of the Hubble tension: early-universe (Planck CMB, arXiv: 1807.06209, H0=67.4+/-0.5) vs late-universe distance-ladder (SH0ES, arXiv:2112.04510, H0=73.04+/-1.04) measurements, ~5 sigma dis... | Report generated: /Users/nramachandra/hep-genesis-testing/projects/ramp-features/reports/261015_224313/report.pdf Title: The Hubble Tension: A Conceptual Summary of Early- and Late-Universe Determinat... |
| 2 | genesis write_report | {"focus": "Conceptual explanation of the Hubble tension: early-universe (Planck CMB, arXiv: 1807.06209, H0=67.4+/-0.5 km/s/Mpc) vs late-universe distance ladder (SH0ES, arXiv:2112.04510, H0=73.04+/-1.04 km/s/Mpc), ~5 sigm... | Report generated: /Users/nramachandra/hep-genesis-testing/projects/ramp-features/reports/261016_170256/report.pdf Title: The Hubble Tension: A Conceptual Summary Prepared Without Computation or Litera... |
| 3 | genesis write_report | {"focus": "Conceptual explanation of the Hubble tension: early-universe (Planck CMB, arXiv: 1807.06209, H0=67.4+/-0.5 km/s/Mpc) vs late-universe distance ladder (SH0ES, arXiv:2112.04510, H0=73.04+/-1.04 km/s/Mpc), ~5 sigm... | Report generated: /Users/nramachandra/hep-genesis-testing/projects/ramp-features/reports/261017_170213/report.pdf Title: The Hubble Tension: A Conceptual Summary Prepared Without Computation or Litera... |